Proposition 8: remaining fixes

12 October 2026 | only outstanding actions | original checklist numbers retained

NOW = needed under the temporary red estimate (R); RED = proves (R); BOTH = shared work. For fixed $A > 0$, (R) requires $a_N(A) \geq 0$, independent of j, tending to zero, with $\|r_{N,j}\|_2 \leq |\delta| a_N(A)$ for all $j \in I_N(A)$.

Notation: $a = h_{min}$, $b = h_{max}$, $\delta = \delta_{ij}^{r,N}$, $L = b^{1-r}$, $J = [a,b]$.

7. Finish the remainder inequalities. [BOTH]

Use $|\tau_{N,j}|$ in the quotient at line 818 for nonzero increments; handle $j=i_N$ separately. Define M_0 to cover the weighted derivative in line 840 as well as the derivative itself (item 10).

Source: thesis lines 818; 826; 840.

8. Complete the red-remainder proof. [RED]

At line 726 replace the integrand by **[dΨ(u+th)-dΨ(u)][h]**. State $\Psi \in C^1$ and that the entire segment lies in U. At line 829 retain $\mathbf{C_g M_0 |\delta|^2}$; after factoring $|\delta|$, another $|\delta|$ remains. Define μ on J and choose $a_N(A)$ independently of j as below. Justify uniform grid constants. In Lemma 7, order the interval endpoints at line 749 and change the enlarged-supremum equality at line 750 to ≤.

$$\mu(s) = \sup\{\|\Psi'(h)-\Psi'(k)\|_2 : h,k \in J, |h-k| \leq s\}.$$

$$C_g = \tfrac{1}{2}(1-r)\max\{a^{1-2r}, b^{1-2r}\}.$$

$$a_N(A) = L\mu(LA/\sqrt{\lambda_N}) + C_g M_0 A/\sqrt{\lambda_N} \to 0.$$

Source: thesis lines 726-728; 749-750; 822-830; Zeidler [1]; DLMF MVT/Taylor [2].

10. Prove derivative continuity and energy properties. [BOTH]

Show $h \mapsto \Psi'(h)$ is continuous in L^2 on J, using its Fourier formula, Gaussian domination and Parseval. The Fourier envelope $b\omega^2 \exp(-a^2\omega^2/2)$ is square integrable. Define M_0 below to bound both $\|\Psi'(h)\|_2$ and $h^{1-r}\|\Psi'(h)\|_2$. Prove Q_r is continuous and positive using $P_n(0)=1$. Continuity also yields $\mu(s) \to 0$.

$$M_0 = \max\{1,L\} \sup_{h \in J} \|\Psi'(h)\|_2 < \infty.$$

Source: thesis lines 358-366; 826; 840; 859; Hunter Thm. 4.24 [3].

11. Finish the Riemann-sum details. [NOW]

State $\varepsilon_N \to 0$ and define $m_N = \lfloor A/\varepsilon_N \rfloor$. Your widths at line 876 use right endpoints: use the rectangle interval $[-(m_N+1)\varepsilon_N, m_N\varepsilon_N]$, whose endpoints tend to -A,A.

$$\varepsilon_N = ((b^r-a^r)/r)\sqrt{\lambda_N}/(N-1) \to 0.$$

Source: thesis lines 782; 876-879; DLMF Eq. (1.4.18) [2].

13. Make notation consistent. [BOTH]

Use one δ and τ notation. Fix $r_{N.j}$, η_{Nj}, ρ_{Nj}, m_n and unindexed ε. Use $S^{(N)}$ consistently. Fix the generic δ in the endpoint identity at line 872, and use $r_u(h)$ in the quotient at line 722.

Source: thesis lines 722; 813-856; 872-879.

14. Qualify the interpretation and prove the Perron claim. [NOW / SEPARATE RESULT]

State the row-sum comparison asymptotically for fixed interior limiting bandwidths. Use Q_r, or explain $\beta=2(1-r)$. Slow change refers to $v=h^r/r$; Q_r measures squared speed. The heading's Perron-concentration claim still needs a separate theorem.

Source: thesis lines 714; 894-903.

All corrections are original derivations from the thesis. Sources for the general tools:

[1] Zeidler (1995), Applied Functional Analysis, §4.4, Prop. 4(ii), p. 241; Cor. 5, p. 242; Example 3, p. 240.

[2] NIST DLMF §1.4, Eqs. (1.4.11), (1.4.18), and §1.4(vi).

[3] Hunter, Measure Theory, Thm. 4.24, p. 49.

Thesis: /Users/iljasukhanov/MASTERARBEIT/Abschlussarbeit_2021/LaTeX/chapters/5_bandwidth_selection.tex